

AI Career Pulse: Decoding the Job Market

Section 2: Data Collection and Preprocessing Phase

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2. Data Collection and Preprocessing Phase

2.1 Data Source

The primary data source is a single flat table, `ai_jobs_market_2025_2026`, connected directly inside Tableau (visible in the Data pane as “`ai_jobs_market_2025...`”). It contains one row per job posting, with categorical, numeric, and boolean fields describing the role, company, and compensation.

2.2 Dataset Scale

- Total job postings analyzed: 91
- Postings by year: 2025 → 36 postings, 2026 → 55 postings (a ~53% year-over-year increase)
- Distinct job titles: 23 (e.g., Senior ML Engineer, AI Solutions Architect, AI Research Scientist, Prompt Engineer, Generative AI Engineer, Robotics Engineer (AI), Computer Vision Engineer, etc.)
- Job categories: 11 (Architecture, Governance, Business, Security, Research, ML Operations, Data Engineering, Robotics, Product, AI Engineering, Data Science)
- Industries covered: 12 (Automotive, Consulting, Education, Energy, Finance, Government, Healthcare, Manufacturing, Media, Research, Retail, Technology)
- Company size bands: 5 (Startup 1–50, SME 51–500, Mid-size 501–5000, Enterprise 5000+, Big Tech / FAANG+)
- Experience levels: 4 (Entry 0–2 yrs, Mid 3–5 yrs, Senior 6–9 yrs, Lead 10+ yrs)

2.3 Raw Fields (as found in the Data pane)

Field Name	Description
Job Id	Unique identifier per job posting (used with COUNTD for distinct counts)
Job Title	Specific role name, e.g., "Senior ML Engineer", "Prompt Engineer"
Required Skills	Text field listing skills required for the posting
Posting Month / Posting Year	Temporal fields used for trend analysis
Remote Work	Indicates remote / hybrid / on-site status
Salary Tier	Pre-binned salary band, e.g., "Upper-Mid (\$150–200k)"
Annual Salary USD	Full annual compensation in USD
Salary Min USD / Salary Max USD	Lower and upper bound of the posted salary range
Years Of Experience	Numeric years of experience required
Benefits Score 10	A normalized 0–10 score representing benefits generosity
Demand Score	A normalized score representing hiring demand intensity for the role

Demand Growth YoY	Year-over-year percentage growth in demand for the role
Is Llm Role / Is Remote Friendly / Is Senior	Boolean/flag fields for role classification
Latitude / Longitude (generated)	Auto-generated geographic fields (for potential future map visuals)

2.4 Data Cleaning Steps

- Verified there were no duplicate Job Id values that would inflate COUNTD-based counts.
- Standardized categorical text fields (Job Category, Industry, Company Size) to consistent casing/spelling so that Tableau does not split a single real-world category into multiple bars due to typos.
- Confirmed numeric fields (Annual Salary USD, Salary Min USD, Salary Max USD, Years Of Experience) were correctly typed as numbers, not strings, so aggregations (SUM/AVG/MIN) compute correctly.
- Reviewed Posting Month / Posting Year for consistent date formatting to support the year-over-year trend view.
- Spot-checked outlier salary values against their Salary Min/Max range to catch any data-entry errors before they skewed averages.

2.5 Calculated Fields Created

Calculated Field	Purpose
Avg Salary Range	Derives a representative mid-point / range from Salary Min USD and Salary Max USD for comparison across roles
AI Salary Premium Pct	Calculates the % salary premium an AI-specific role commands versus a baseline, highlighting the AI skill premium
Demand Score / Demand Growth Yoy	Quantifies and tracks how hiring demand for a role is changing over time
Is Llm Role (Count)	Flags whether a posting is for an LLM-specific role, enabling LLM-vs-non-LLM comparisons
Is Remote Friendly	Flags remote-eligible postings for remote-work analysis
Is Senior	Flags postings requiring senior-level experience
Job Id (Count (Distinct))	A COUNTD(Job Id) measure used throughout the dashboard to count unique postings rather than duplicate rows

2.6 Salary Tier Binning

Raw Annual Salary USD values were grouped into a Salary Tier dimension (e.g., "Upper-Mid (\$150–200k)") so that comparisons across industries, company sizes, and job titles could be made at a band level rather than requiring a continuous axis — this makes cross-category comparison in bar charts far more readable.

2.7 Filtering Strategy

Three filters were applied globally at the data-source level so that every worksheet (and therefore the dashboard) stays in sync:

- Country — allows drill-down into a specific country or a global (All) view.

- Education Required — allows filtering by minimum qualification (e.g., Associate's, Bachelor's, Master's).
- Salary Tier — allows the viewer to focus on a specific compensation band, e.g., Upper-Mid (\$150–200k), which was used as the primary lens for this analysis.

In addition, Top-N context filters were layered on top of individual views to keep charts focused on the most significant categories:

- Top 5 by COUNTD([Salary Tier]) — restricts a view to the five most common salary tiers.
- Top 5 by COUNTD([Job Id]) — restricts a view to the five job titles/categories with the highest posting counts.

2.8 Challenges During Preprocessing

- Balancing SUM vs. AVG aggregations early on — total salary by category (SUM) tells a very different story than average salary by title (AVG), so both needed distinct calculated logic.
- Ensuring the global filters (Country, Education Required, Salary Tier) didn't conflict with each sheet's own local Top-N context filters.